Pointless Coherence on Glued Carriers

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> Answers can be consistent without being about anything, as long as the contradiction hides in a question nobody can ask.

This note follows that sentence through a sandwich shop and Penrose's impossible staircase, and out the other side into a theory. The puzzles come first, because the theory is only a careful account of what they show. Precise statements sit in short *Precisely* asides and in the appendix, where proofs and status labels live ([Lean] machine-checked, [hand] my proof, [cited] literature).

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1 The sandwich shop

A sandwich shop has six sandwiches on its menu. Each is made of exactly *two* of four fillings, so every pair of fillings appears once:

sandwich	cheese	ham	egg	tomato
cheese & ham	✓	✓		
cheese & egg	✓		✓	
cheese & tomato	✓			✓
ham & egg		✓	✓	
ham & tomato		✓		✓
egg & tomato			✓	✓

The online order form asks about one filling at a time: *Cheese? yes/no. Ham? yes/no. Egg? yes/no. Tomato? yes/no.* The owners wrote the form's validation and know nothing about the menu. Their only rule: **every line must be answered, with exactly one of yes or no.**

A picky customer submits *no, no, no, no.* Does it pass validation?

line	yes	no	ticks
Cheese?		✓	1 ✓
Ham?		✓	1 ✓
Egg?		✓	1 ✓
Tomato?		✓	1 ✓

Every line has exactly one tick, so the order is *consistent*: it passes the only check the owners know how to make.

Can the kitchen make it? Say a sandwich *fits* the order if every answer on the form is right for that sandwich. Hold each sandwich up against the order:

	cheese?	ham?	egg?	tomato?	fits?
the order	**no**	**no**	**no**	**no**	
cheese & ham	yes	yes	no	no	×
cheese & egg	yes	no	yes	no	×
cheese & tomato	yes	no	no	yes	×
ham & egg	no	yes	yes	no	×
ham & tomato	no	yes	no	yes	×
egg & tomato	no	no	yes	yes	×

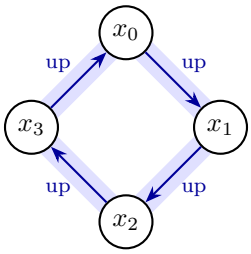
Every sandwich has two fillings, so nothing on the menu fits. A valid order the kitchen cannot make.

Where did the contradiction go? Each line on its own was fine. Even "no cheese, no ham" is fine, since egg & tomato fits. The trouble appears only when all the "no"s are combined, and the form has no line for combinations. "No cheese *and* no ham *and* no egg?" is not a question on this form. The contradiction lives in a question that is never asked.

This is the whole phenomenon in miniature. Here it is harmless: there are only six sandwiches, nothing is infinite, and the cure is simply to add combination lines to the form.[1]

2 The impossible staircase

Penrose's staircase is a drawing of four flights of stairs joined in a loop. Each flight climbs to the next landing, and after four flights you are back where you started.



The drawing has everything the theory is about:

- *every local view is coherent:* any one flight, looked at alone, is a perfectly good picture of real stairs;
- *neighbouring views agree:* where two flights meet, they show the same landing;

[1] Mathematically this is an example of Navara and Pták, *Čas. pěst. mat.* 108 (1983): six possibilities and four yes/no questions, each possibility answering "yes" to exactly two.

- *there is no whole:* no building is what the drawing depicts;
- *the contradiction is only in the whole:* no single flight shows it. You see it only by following the entire loop.

To make this mathematics, give the landings heights x_0, x_1, x_2, x_3, each a whole number. A *view* is one flight. It shows two neighbouring landings, and it answers any yes-or-no question about their two heights: "is the next landing higher?", "are the heights 30 and 70?". It also answers the opposite of any question, and questions that lump possibilities together, even endlessly many: "is x_0 even?". Answers are *consistent* if they hang together in two ways. First, split "is x_0 at most 10?" into "at most 5?" and "6 to 10?". The three answers must fit one of these rows:

at most 10?	at most 5?	6 to 10?
yes	yes	no
yes	no	yes
no	no	no

Never two yeses, and a yes to the whole needs a yes to one of its parts. The same goes for any split into a few non-overlapping parts. Second, two flights that share a landing say the same things about it. Every flight in Penrose's drawing says *the next landing is higher*.

A straight staircase is always buildable. Unbend the loop into a straight run of three flights, x_0 to x_1 to x_2 to x_3, each going up. Real heights $10, 20, 30, 40$ satisfy every view: a real building.

The loop is not. Try real heights against the four views:

heights (x_0, x_1, x_2, x_3)	x_1 higher?	x_2 higher?	x_3 higher?	x_0 higher?	fits?
the drawing	**yes**	**yes**	**yes**	**yes**	
$10, 20, 30, 40$	yes	yes	yes	no	×
$50, 90, 20, 70$	yes	no	yes	no	×
$30, 30, 30, 30$	no	no	no	no	×

Going round, the heights must come back down somewhere. With real heights the views cannot even agree with each other: fixing actual heights on each flight forces one global set of heights, and that set fails. So with real numbers the drawing is not merely unbuildable; its views are inconsistent.

3 Heights beyond every number

The drawing becomes a genuinely consistent set of views once the heights are allowed to be *ideal*. This step has no everyday counterpart: an ideal height is not a height anything has. It is a complete, consistent rule for answering every question about one number as if that number lay beyond every number you could name.

- "Is $x_0 = 5$?" No. "Is $x_0 = 500$?" No. No to every particular number.
- "Is x_0 bigger than any number you name?" Yes.
- "Is x_0 even?" A definite yes or no, fixed once and for all.

Precisely: an ideal height is a non-principal ultrafilter p on $\mathbb{N}$. Its existence needs a little of the axiom of choice.

A flight answers questions about its two landings by taking both from beyond every number, the upper one *further out* than the lower. "Is the next landing higher?" Yes: whatever the lower height is, almost every further-out choice beats it.

Precisely: "the upper one further out" is the ordered product $p \otimes p$: a question A about (x_0, x_1) gets "yes" when p says yes to the set of x_0 for which p says yes to $\{x_1 : (x_0, x_1) \in A\}$. Its two marginals are both p, and it is not symmetric.

On a straight staircase this always fits together. One ideal description of the whole run, "x_0 beyond every number, x_1 further out, then x_2, then x_3", answers every question about all four landings at once. Each flight's answers are just its answers about two of them. Call a description of all the landings together an *ideal building*. A real building is an ideal building too, the trivial one.

Precisely: an ideal building is an ultrafilter on the whole space of possibilities, an *ideal point*; a real building is a *point*.

On the loop, every check passes and still nothing is built. Let every flight of Penrose's loop use ideal heights, upper landing further out. Every flight genuinely says "up". Two flights that share a landing agree about it, since both describe it by the same ideal rule p. The checks:

check	what the views say	passes?
each flight, alone	"next landing higher": yes; "height n": no for every n	✓
landing x_0, seen from both of its flights	the same answers (both by p)	✓
every question built from the views, and its opposite	exactly one yes [Lean]	✓
some real building fits	no: see the table in §2	×
some ideal building fits	no: see below	×

The third row is the theorem, checked in Lean: *every* question that can be built from the views, including combined questions spanning neighbouring flights, gets consistent answers. And the question that would expose the loop, "is x_0 below x_1 below x_2?", is not among them. No view shows three landings, just as no patch of the drawing shows the whole loop.

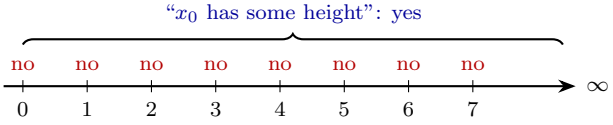
There can be no ideal building. An ideal building answers every question about all four landings, the forbidden one included. It would have to say $x_0 < x_1 < x_2 < x_3 < x_0$.

So the views are consistent and nothing fits them: no real building, and not even an ideal one. Call such answers **pointless**. Each flight's landings are ideal; the loop as a whole is no building at all.

The sandwich order was pointless too, but trivially so, because the menu was finite. The staircase is pointless and infinite, and that is what makes it sharp.

4 What infinity demands

Ask the staircase "what is the height of x_0?", and split the question into its endlessly many options: $0, 1, 2, \ldots$. Every option gets "no". The whole question, "x_0 has some height", gets "yes".



Consistency only covers splits into a few parts, and those are fine: "at most 100?" no, "above 100?" yes. Now demand more: the "yes"es must add up across *endlessly* many options too.

This is σ-additivity, the usual rule of probability theory. Under it some single height must get the yes. Every landing is pinned to a real height, and real heights cannot climb around a loop.

On these boards, demanding that answers add up over infinitely many options is the same as demanding that they come from a real building (Proposition A.4). The demand for a completed answer is exactly what makes the staircase impossible.

5 Why loops: the first ideas of the theory

The sandwich shop and the staircase show the whole mechanism. Separating its parts gives the theory's first ideas. From here on the staircase's words give way to general ones:

staircase	**general word**
a landing's height	a *slot*
a flight (one view)	a *window*
a flight's way of answering	a *recipe*
a real building	a *point*
an ideal building	an *ideal point*
which flights join which landings	the *window graph*

Contradictions need questions that cannot be asked. Consistent answers respect every "and" that is actually on the board: if two questions and their joint question can all be asked, two yeses force a yes on the joint question (Lemma A.3). So a pattern can fail to be fitted only through joint questions the board lacks. That is the space pointlessness lives in.

No triangle keeps things local. Questions put to windows that share no slot always overlap, since they constrain different slots. So when a question is split among several windows, those windows must pairwise share a slot. If the window graph has no triangle, pairwise-touching windows always share one *common* slot: they form a star around it. Slicing at that slot shows that the star's questions stay a star's questions under "not". The result is that every buildable question is a question to one star of windows (Theorem A.5), so consistent answers are completely determined by what each window says. This covers trees, every ring of length at least four, and every other triangle-free graph. A triangle is the one shape where three windows touch pairwise with no common slot, and there locality fails. (In the language of hypergraphs: the windows have the *Helly property*.)

Trees always glue. Two neighbouring windows' ideal recipes, agreeing on their shared slot, always combine into one recipe for all three slots (Lemma A.6). On a tree you can keep combining, one window at a time, so on any tree, line or branching, every consistent pattern is an ideal point (Theorem A.7). Nothing pointless can happen without a loop.

Loops can carry a twist. Around a loop the recipes may fail to combine, as the staircase's directions do. Call a family of window recipes *flat* if it combines into one recipe for everything, and *twisted* otherwise. On triangle-free boards, a pattern is an ideal point exactly when its window family is flat (Proposition A.9). Pointless patterns are twisted families that still extend consistently to every buildable question.

Loops travel. Suppose a board contains a loop of four or more windows with no shortcut across it: no window joins two slots of the loop that are not neighbours on it. Fix every slot off the loop at some number. As far as the board's questions can tell, it is now the ring. A question to a window touching the loop becomes a question about one loop slot, and a question off the loop becomes trivially yes or no. So the staircase's answers carry over to the bigger board, and they are still pointless there (Theorem A.8). Any board with such a loop has pointless patterns, whatever else it contains.

Infinity forbids the twist. Under σ-additivity every window is pinned to actual numbers, and actual numbers carry no direction, so there is nothing to twist.

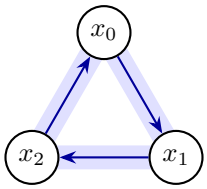
The map so far. Putting the boards together:

	tree	**a shortcut-free loop of 4 or more**	**every loop has a shortcut** (triangles)
finitely many values	actual	actual	actual
infinitely many values, answers add up over infinitely many options	actual	actual	actual
infinitely many values, finite adding only	ideal [hand]	**pointless exists** [Lean ring + hand transport]	**pointless exists** [Lean triangle + hand transport]

In graph language the columns are: forests; graphs with an induced cycle of length ≥ 4 (non-chordal); and chordal graphs that are not forests. The shaded cells are where the theory lives: a loop, infinitely many possibilities, and only finite adding. The last column was the open edge until the triangle was settled (§6); the line now falls between trees and everything else.

6 The triangle

Now use only *three* flights: windows $\{x_0, x_1\}$, $\{x_1, x_2\}$, $\{x_2, x_0\}$, every one going up. This is Penrose's impossible *triangle*, the tribar, and Specker's seer with three boxes: any two flights are fine, all three together are impossible for real heights.



Now every two flights share a landing, but no landing is on all three flights: in window terms every two windows share a slot, but no slot is shared by all three: the window graph is a triangle, the one shape the locality argument cannot handle. Questions can braid all three windows. With two possible values, "are all three equal?" becomes askable, though it cannot be broken into window questions. With three possible values that question disappears, and 2,088 others take its place. The ring's method, reading every answer off neighbouring windows, no longer applies.

So the triangle was the experiment that decided what the theory is about. Every loop either has a shortcut-free stretch of four or more, which is already settled, or is made of triangles.

- *If the triangle carried pointless answers,* then so would every board with any loop. Triangles travel too, since a triangle never has a shortcut. Pointless coherence would then happen *exactly* on boards that are not trees.
- *If it did not,* the expected line was *chordal* versus *non-chordal*: boards whose every long loop has a shortcut versus those with a shortcut-free one.

Either answer matches a classical pattern. Trees are where pairwise probability tables always fit together (Wainwright–Jordan). Chordal graphs are where junction trees and Vorob'ev's theorem live. It carries them. The rest of this section is the puzzle that shows why.

Forced answers. Any consistent answers extending the three flights' views are *forced* on every question built in finitely many steps: the answer must be the sum of the flights' limits (Proposition A.10). Three landings taken from beyond every number can be drawn in six orders, and each order gives a genuine ideal building. Up to the sign of one building, the forced answer is the sum of what three of those buildings say. It fails to be 0 or 1 only on a question that holds in exactly the three orders that go round the loop the staircase's way: the tribar's own twist. Call that the *cyclic question*: "do the three heights go round the staircase's way?" (Proposition A.13). So everything turns on one thing. Can the cyclic question, or anything that looks like it far out, be built?

The balance rule. Pick two candidate heights for each landing. That gives $2 \times 2 \times 2 = 8$ buildings, the corners of a box. Mark a corner $+$ or $-$ by whether it uses an even or odd number of second choices. A window question answers each corner by looking at two landings only. The corners pair up across the third landing, one $+$ and one $-$ in each pair, so *every window question says yes equally often on the $+$ and $-$ corners.* "Not" keeps the balance, since every box has four of each, and so does putting together questions that never both say yes. So every buildable question, built in any number of steps, *balances on every box.*

The lift. Take heights $1, 4, 7$ for x_0, x_1, x_2: they go up round the loop, $x_0 < x_1 < x_2$, the staircase's way. Now lift the lowest landing above the highest: $x_0 = 10$. The heights $10, 4, 7$ still go round the staircase's way ($x_1 < x_2 < x_0$). Put both into one box, with second choices 10 for x_0, 8 for x_1 and 2 for x_2:

corner (x_0, x_1, x_2)	**sign**	**way round**
$(1, 4, 7)$	$+$	staircase's way
$(10, 8, 7)$	$+$	the other way
$(10, 4, 2)$	$+$	the other way
$(1, 8, 2)$	$+$	the other way
$(10, 4, 7)$	$-$	staircase's way
$(1, 8, 7)$	$-$	the other way
$(1, 4, 2)$	$-$	the other way
$(10, 8, 2)$	$-$	the other way

Suppose a buildable question says no to every configuration that goes the other way. Then the balance rule leaves one corner on each side, so it answers $(1, 4, 7)$ and $(10, 4, 7)$ alike. Lifting the lowest landing never changes its answer. Enough lifts carry any spread-out configuration that goes the staircase's way to any other, so such a question says yes to all of them or to none.

Why that settles it. Try to build the cyclic question.

- "Not" turns it into the question that goes the other way, which the reflection $x_1 \leftrightarrow x_2$ turns back.

- Putting together pieces that never both say yes: each piece says no to the other way, so by the lift each piece takes all of the spread-out staircase-way configurations or none of them. One piece takes them all. It already looks like the cyclic question far out, and it was built earlier.

So nothing that looks like the cyclic question far out is ever built, not even in infinitely many steps (Theorem A.14). Ramsey's theorem, run along the ultrafilter, finishes the job. Any question whose forced answer is 2 looks like the cyclic question on some infinite set of heights, and two balance-rule boxes pin its pattern there. So the forced answers are always 0 or 1, and *the triangle carries pointless answers* (Theorem A.15, machine-checked).

The dividing line. With the transport of loops (Theorem A.8): *a board carries pointless answers exactly when its window graph is not a tree* (not a forest). This is for two-slot windows and infinitely many values. *Credit (audit, October 2026):* the dividing line and the transport method (reduce any cycle to a triangle or an induced cycle of length ≥ 4, then fix the other coordinates) are those of Atserias–Kolaitis, arXiv:2009.09488, Lemmas 13–15, for relations and bags; what remains here is the two base witnesses at the level of two-valued finitely additive states, and the locality step on the full carrier [unclear: owners unread]. With finitely many values every consistent pattern is actual (Proposition A.4), and a window that sees all three landings at once makes the triangle an ordinary board.

What the theory must settle.

1. *Extension on triangle-free graphs.* There, a consistent pattern is the same as a family of window recipes that extends, and pointless means twisted (Theorem A.5, Proposition A.9). Which twisted families extend? The rings and the triangle say some do. Is it all of them, on every loop?
2. *The classification.* Which twisted families extend to every buildable question? The hope is that twist is measured by something like a winding number of the directions around each loop. On the triangle the balance rule and the lift are the first instruments.
3. *Wider windows.* Windows showing three or more slots at once.
4. *No background at all.* The boards assume a space of *possibilities* (the slots and their possible values) and nothing more. Drop even that, keeping only the questions. Does pointlessness survive when there is no space to lack a point in?

7 Coda: where this sits

Its neighbours. Each neighbouring theory covers a region and stops at the edge of the shaded cells.

- *Lines.* Vorob'ev (1962) and database theory show that consistent data on line-like structures always comes from a whole, for ordinary probabilities.
- *Loops, with ordinary probabilities and finitely many outcomes.* The sheaf-theoretic account of contextuality owns this: Abramsky and Brandenburger, and for loops exactly, Kharoof, Ipek and Okay (2024).
- *Infinity on ordinary (Boolean) boards.* De Finetti and Howson. On a board where every "and" can be asked, consistent answers are always ideal points.
- *Glued boards with answers that add over infinitely many options.* The Czech school of quantum logic: Navara and Pták (1983) and Dravécký and Šipoš (1980, Example 6). The latter applies to the triangle's *flights* (the differences x_1-x_0, x_2-x_1, x_0-x_2), which by themselves already form a closed board with pointless answers. Those answers do not extend to the full triangle board, where adding over infinitely many options forces

a point. The triangle theorem is exactly that extension, which is possible only with finite adding.

None of them covers finitely additive answers on the full glued board. That gap is where this theory sits.

Standard names. In the literature's terms: a *state* is a two-valued finitely additive state; a *point* is a Dirac (concentrated) state; an *ideal point* is a coherent state, one that extends to all sets (Bhaskara Rao and Bhaskara Rao, Thm. 3.2.10); *pointless* means non-coherent; a *board* or *carrier* is the σ-class generated by coordinate sub-σ-algebras; the *window graph* is the marginal scheme (measurement cover). The puzzle words are kept for the story; the theorems are stated so that they can be read in either vocabulary.

Its lineage. The picture is old.

- *Specker* opened the paper that became the Kochen–Specker theorem (1960) with a seer's three boxes. Any two opened hold exactly one gem, which is pairwise consistent and jointly impossible. That is the triangle.
- *Penrose* explained the impossible staircase and tribar by the cohomology of a loop (1991).
- *Aristotle* held that the infinite exists only potentially, never as a completed whole. Finite adding is the potential infinite. Adding up over infinitely many options is the actual one, and the staircase is a case where demanding the actual infinite excludes what the potential allows.
- *Kant* called the completed series of conditions not *given* but *aufgegeben*, set as a task: "nicht als Axiom, die Totalität im Objekt als wirklich zu denken, sondern als ein Problem für den Verstand". The staircase is a small antinomy. Every window is right, and the totality read as an object contradicts them, while read as a task it does not.

What the results say. The theory studies one gap: answers that are *complete* (every askable question answered) and *coherent*, yet about no world. In ordinary logic the completeness theorem closes that gap, because every "and" can be asked. On glued boards it closes in three regimes and opens in one:

regime	**do coherent answers earn a world?**
finitely many values	yes, an actual one
infinitely many, adding over the whole infinity	yes, an actual one
infinitely many, finite adding, no loop	yes, an ideal one
infinitely many, finite adding, a loop	**no**: pointless answers exist

Three further findings shape where the theory goes next.

- *The board contributes one unaskable question.* On the triangle it is "do the heights go round?"; by the balance rule and the lift it can never be built. Beyond that, whether answers are pointless depends only on whether the windows' answers glue together. Extension was never the obstacle in any case checked.
- *Renaming and motion never create impossibility.* A window that only renames a landing can be contracted away, and symmetry only moves worlds around. The twist lives in the loop, not in the action.
- *Only "beyond" goes round.* A flight that draws one landing beyond the other carries a direction. A loop of such directions is a task that would have to finish before it started. For the simplest window types (independent draws and renamings) the twist is exactly such a directed cycle [hand]. In general it is not. Beyonds also *compose*: a

triangle whose directions never go round can still fail to close, because the combined step $x_2 - x_0$ must equal the composite of the two others in the algebra of ultrafilters, and one window can ask for a different composite [hand]. The invariant is a composition law, with orientation as its shadow.

In these terms Aristotle's potential infinite is finite adding, and Kant's totality *set as a task* is the ideal point. A loop is where the task becomes circular.

Its stance. Throughout, answers come first. Nothing is assumed to be giving them. Only the space of possibilities is assumed. Points, actual or ideal, are what we get when answers can be reconstructed into a whole. The theory is the study of when that reconstruction succeeds, and of what remains when it fails.

A Definitions and results

Definition A.1 (boards and answers)**.** A *possibility space* is $\Omega = X^V$, with V finite and X an alphabet. A *window* is a two-element $\{i, j\} \subseteq V$, and the windows form a graph. A *window question* is $\{x : (x_i, x_j) \in A\}$ with $A \subseteq X^2$ arbitrary. The *carrier* $\mathcal{D}$ is the smallest family containing all window questions and closed under complement and countable disjoint union: the buildable questions. A *state* (consistent answers) is $s : \mathcal{D} \to \{0, 1\}$ with $s(\Omega) = 1$ and $s(A \sqcup B) = s(A) + s(B)$ for disjoint A, B. It is *σ-additive* if additivity holds for countable disjoint unions.

Definition A.2 (points)**.** A state is a *point* (fitted by an actual assignment) if $s(E) = [\omega \in E]$ for some $\omega \in \Omega$. It is an *ideal point* if $s(E) = [E \in W]$ for an ultrafilter W on Ω. It is *pointless* if it is neither.

Lemma A.3. *[hand; standard] Let s be a state on $\mathcal{D}$.*

(a) If A, B and $A \cap B$ are in $\mathcal{D}$ and $s(A) = s(B) = 1$, then $s(A \cap B) = 1$.

(b) s is an ideal point iff its true sets have the finite intersection property.

Proof. (a) $A \setminus B = (A \cap B^c)$ lies in $\mathcal{D}$, being A minus $A \cap B$, and lies inside B^c. So $s(A \setminus B) \leq s(B^c) = 0$.

(b) Extend the true sets to an ultrafilter; the false sets have true complements.

□

Proposition A.4. *[Lean: ring, `no_sigma_extension`; hand: general] If X is countable, every σ-additive state on a glued carrier is a point. If X is finite, every state is a point.*

Proof.

- On each window the fibres $\{(x_i, x_j) = (a, b)\}$ form a countable (or finite) partition, so exactly one is true.
- Shared coordinates make these choices agree, giving one ω.
- The sets on which s agrees with δ_ω form a carrier containing the window questions, hence all of $\mathcal{D}$.

□

Theorem A.5 (no triangle: locality)**.** *[hand; machine-checked on finite alphabets] Suppose the window graph has no triangle; equivalently, the windows form a Helly family. Then every $E \in \mathcal{D}$ is a disjoint union $\bigsqcup_i E_i$ of window questions on windows that all contain one common coordinate (a star). So every state satisfies $s(E) = \sum_i u_{w_i}(E_i)$ and is determined by its window ultrafilters.*

Proof.
- *Rigidity.* Window questions on windows with no shared coordinate meet if both are non-empty. So in a disjoint union of window questions, the non-empty pieces lie on pairwise adjacent windows. Without triangles, these form a star at some coordinate c.
- *Slicing.* Fix $x_c = a$. Each piece's slice is a cylinder in its own leaf coordinate; they are pairwise disjoint in independent coordinates, so at most one is non-empty. The slice of E^c is then a cylinder in the same coordinate, or everything. Gathering slices by leaf writes E^c as a star at c.
- *Closure.* Let $\mathcal{G}$ be the disjoint unions of one window question per window. It is closed under complement (by slicing) and under countable disjoint union (regroup window by window). So $\mathcal{G} = \mathcal{D}$.

Machine check of $\mathcal{D} = \mathcal{G}$ (script `glued_helly_check.py` in the verification folder):
- holds on the star $K_{1,3}$ (binary and ternary), a branching tree, the 4- and 5-cycles, and $K_{2,3}$;
- fails whenever there is a triangle: triangle 70 vs 66; triangle with pendant 98 vs 94; 4-cycle with chord 148 vs 132.

Compare Greechie's loop lemma (Kalmbach 1983, §4 L9). □

Lemma A.6 (gluing)**.** *[hand; very likely folklore] Ultrafilters on $X \times Y$ and $Y \times Z$ with the same Y-marginal are the projections of one ultrafilter on $X \times Y \times Z$.*

Proof. Members of the two ultrafilters project into the common marginal, so they meet; the sets $A \times Z$ and $X \times B$ have the finite intersection property. □

Theorem A.7 (trees)**.** *[hand] If the window graph is a forest, every state is an ideal point.*

Proof.
- Order each tree's windows so that each meets the earlier ones in one coordinate.
- Glue the window ultrafilters one at a time (Lemma A.6). Join separate trees by the product of their ultrafilters.
- This gives W projecting to every u_w.
- By Theorem A.5, $s(E) = \sum_i u_{w_i}(E_i) = \sum_i W(E_i) = W(E)$.

□

Theorem A.8 (loops travel)**.** *[hand, from the Lean ring witness] If the window graph contains an induced cycle of length $L \geq 4$, there is a pointless state over $\mathbb{N}$ with no σ-additive version. If the triangle carries a pointless state, so does every graph containing a triangle.*

Proof.
- Let C be the induced cycle. Define $\psi : \mathbb{N}^C \to \mathbb{N}^V$ by copying the cycle coordinates and setting every other coordinate to a constant.
- Pull back each window question of the graph:
  - a window on the cycle gives a window question of the ring, since the cycle is induced, so there is no chord;
  - a window from the cycle to outside gives a question about one ring coordinate;
  - a window off the cycle gives $\emptyset$ or everything.
- The sets whose preimage is in $\mathcal{D}_C$ form a carrier, so $\psi^{-1}(\mathcal{D}_G) \subseteq \mathcal{D}_C$. Hence $s = \mu \circ \psi^{-1}$ is a state, with μ the ring witness (Theorem A.16).
- *Pointless.* An ultrafilter fitting s would push forward, along the projection to the cycle coordinates, to one fitting the staircase on the ring.
- *No σ-additive version:* Proposition A.4.
- *The triangle clause:* the same argument, since a triangle is always induced.

The Lean development already proves the analogous transport along cycle maps, in the lemmas `pullback` and `sL_sigmaEssential`. This graph form is not yet in Lean. □

Proposition A.9 (flatness)**.** *[hand] On a triangle-free window graph, a state is an ideal point iff its window ultrafilters glue to one ultrafilter on Ω (a flat family).*

Proposition A.10 (the forced functional)**.** *[hand] On the triangle (and on rings), every buildable question built in finitely many steps has indicator $f_0(x_0,x_1)+f_1(x_1,x_2)+f_2(x_2,x_0)$. Any state extending a compatible family (u_k) takes the value $\Lambda = \sum_k \lim_{u_k} f_k$ there. Λ is well defined: if $f_0+f_1+f_2=0$, each f_k is $\varphi_i(x_i)-\varphi_j(x_j)$ plus a constant, the constants summing to 0, and these terms cancel because the marginals agree. So a family can extend only if $\Lambda \in \{0,1\}$ on every finitely built question.*

Proposition A.11 (cut the loop)**.** *[hand] On a triangle or ring with window ultrafilters $u_k = p \otimes p$ in a cyclic orientation, fix a window k and let $W^{(k)}$ be the ultrafilter gluing the other windows along the path left after cutting k. For every E whose indicator is a finite sum $\sum_j f_j$ of bounded window functions,*

$$\Lambda(E) = W^{(k)}(E) + \delta_k(E), \qquad \delta_k(E) = \lim_a \lim_b f_k - \lim_b \lim_a f_k,$$

the commutator of the iterated p-limits on window k, in the two orders. δ_k is well defined, since single-coordinate terms have order-independent limits. So $\Lambda(E) \in \{0,1\}$ whenever E's window-k part is zero.

Proof. $W^{(k)}$ projects to u_j on every window $j \neq k$ and to u_k^T on window k. Hence $W^{(k)}(E) = \sum_{j\neq k} \lim_{u_j} f_j + \lim_{u_k^T} f_k$; subtract from $\Lambda(E)$. □

Theorem A.12 (rings, conceptual proof)**.** *[hand; agrees with Lean] For $L \geq 4$, Λ is a two-valued finitely additive state on the whole ring carrier extending the staircase.*

Proof.
- By Theorem A.5, every $E \in \mathcal{D}$ is $E_{k-1} \sqcup E_k$, so 1_E is a sum of two window functions.
- Zero-sum families of window functions on a cycle are differences of single-coordinate functions plus constants, so Λ is well defined and linear.
- Pick a window $m \notin \{k-1, k\}$. Then E's window-m part is zero, so $\Lambda(E) = W^{(m)}(E) \in \{0,1\}$ (Proposition A.11).
- Linearity gives finite additivity, and $\Lambda(\Omega) = 1$.

In words: cut the staircase where the question is not looking, and read off a genuine ideal building. This explains the case analysis behind `mu_cases`; Lean remains the certificate. The argument inherits the locality theorem, which is a hand proof checked on finite alphabets. □

Proposition A.13 (the triangle, finitely built questions)**.** *[hand] On the triangle with the staircase family, $\Lambda(E) \in \{0,1\}$ for every question E built from window questions by finitely many complements and disjoint unions.*

Proof.
- *Six buildings.* For each of the six orders π of (x_0,x_1,x_2), let W_π be the ultrafilter "draw in order π, each further out" (iterated $p \otimes p$). Its projection to window $\{i,j\}$ is $p \otimes p$ taken in the order π gives i,j. The *profile* of E is $\pi \mapsto W_\pi(E)$.
- *Profiles of finitely built questions.* A window question's profile depends only on whether π puts i before j: it is constant or a "half" $\{\pi : i \text{ before } j\}$ or its complement. Complements complement the profile. Disjoint questions lie in no common ultrafilter, so their profiles are disjoint and the union's profile is the union. Among constants and halves, the only disjoint pairs are complementary halves or involve $\emptyset$. So every finitely built E has a constant or half profile.

- Λ *from profiles.* Applying the cut identity at two windows $k \neq j$ gives $\Lambda(E) = W^{(k)}(E) + W^{(j)}(E) - W^{(jk)}(E)$. Here $W^{(k)}, W^{(j)}$ are orders going round the loop the staircase's way (rotations of $0, 1, 2$), and $W^{(jk)}$, which reverses windows j and k, goes the other way (a rotation of $2, 1, 0$). This holds termwise on each window.
- *Check.* On constants and halves this gives 0 or 1. It gives 2 only on the profile "exactly the three rotations of $0, 1, 2$", the staircase's cyclic orientation at infinity, which is not a half.

□

The old ring sweep searched every 0/1 function of the right algebraic shape (a sum of window functions) on a finite tower. Its failure at $L = 3$ lies on such a function with the cyclic profile, which is not buildable. It never obstructed the triangle.

Theorem A.14 (the cyclic question is not buildable)**.** *[Lean: `not_inP_of_mem`, `estar_not_mem`] Let C (resp. A) be the strict cyclic (resp. anticyclic) points of $\mathbb{N}^3$. No set that agrees with C, or with A, on the distinct-coordinate points of P^3 for some infinite $P \subseteq \mathbb{N}$ lies in the triangle carrier. In particular $E^* = \{[x_0 \leq x_1] + [x_1 < x_2] + [x_2 < x_0] = 2\}$ is not buildable, although $1_{E^*} = [x_0 \leq x_1] + [x_1 < x_2] + [x_2 < x_0] - 1$ is a sum of window functions. (Over three values the carrier is exactly the 0/1 sums of window functions, a machine count; the gap is only at infinity.)*

Proof.
- *Balance.* Every carrier set has vanishing triple interaction on every $2 \times 2 \times 2$ box: window events do, and the identity survives complements and countable disjoint unions (pointwise sums).
- *Lift.* Thin P to Q (every other element), so that a P-element lies between any two Q-elements. Let G balance and avoid A on P^3. For $a < b < c < a'$ in Q, choose $\gamma \in P$ between a, b and $\beta \in P$ between c, a'. The box $\{a, a'\} \times \{b, \beta\} \times \{c, \gamma\}$ has its other six corners in $A \cap P^3$, so $1_G(a, b, c) = 1_G(a', b, c)$. Rotations give the other two orders, and lifts connect any two points of $C \cap Q^3$.
- *Induction.* The family is closed under complement (and under the reflection (x_0, x_2, x_1) for the A side) and contains no window event. In a countable disjoint union, exactly one piece takes $C \cap Q^3$, and it is a member of the family built earlier. □

Theorem A.15 (the triangle)**.** *[Lean: `triangle_pointless_state`, `sstar_window01/12/20`] For every non-principal ultrafilter p, $s^* = W_{012} + W_{120} - W_{102}$ is finitely additive with $s^*(\Omega) = 1$, is $\{0, 1\}$-valued on the triangle carrier, and equals the staircase limit on every window question. No ultrafilter on $\mathbb{N}^3$ reproduces it, since $\{x_0 < x_1\} \cap \{x_1 < x_2\} \cap \{x_2 < x_0\} = \emptyset$. So the triangle carries a pointless state.*

Proof.
- s^* is a signed sum of three ultrafilters, so it is finitely additive on every set. On a window question two terms cancel or coincide, leaving the oriented $p \otimes p$ limit.
- Suppose $s^*(E) \in \{2, -1\}$ for a carrier set E. The six order conditions intersect to a $p^{\otimes 3}$-large set, and the ultrafilter proof of Ramsey's theorem gives an infinite P on which E follows its six-order profile.
- Two balance boxes on six elements of P cut the 64 patterns to the 10 in the span of window functions. Of those, $s^* = 2$ only on the cyclic pattern and $s^* = -1$ only on the anticyclic one. Theorem A.14 excludes both. □

That s^* is the forced functional Λ on finitely built questions is Proposition A.10 [hand]. The six-orders argument and the Ramsey step sit near known work on products of ultrafilters (Blass; Hindman–Strauss, unread). They are offered as theory content, not as a novelty claim.

Theorem A.16 (the staircase)**.** *[Lean: `mu_cases`, `sL_sigmaEssential`] For every $L \geq 4$, the window ultrafilters $u_k = p \otimes p$ on the ring extend to a state on the whole carrier over $\mathbb{N}^L$. No ultrafilter on $\mathbb{N}^L$ projects to all u_k, since every $\{x_k < x_{k+1}\}$ would be in it, and their intersection is empty. So the state is pointless, and by Proposition A.4 it has no σ-additive version.*

At $L = 4$ the carrier's centre is literally $\{\emptyset, \Omega\}$ and all six admissibility conjuncts hold (Lean, `D4_six_conjuncts`). No prior-art owner was found in three passes.

B Status, sources, files

Status.

- [Lean]: Theorems A.16, A.14 and A.15; Proposition A.4 on the ring.
- [hand]: everything else in Appendix A, including the assembly "pointless $\iff$ not a forest" (method owned: Atserias–Kolaitis 2020, Lemmas 13–15).
- Open: the classification and extension questions at the end of §6.
- Prior art for the triangle: two scouting passes and a hostile audit; see the taxonomy file below.
- Prior-art risk for the staircase: Bhaskara Rao, *Theory of Charges*; Goldberg on ultrafilter products.

Sources for the coda , with how each was checked:

- read in full:
  - Kant, *KrV*, Antinomy §§7–8 (German, Gutenberg ed.);
  - Specker, *Dialectica* 14 (1960), in Seevinck's translation, arXiv:1103.4537;
  - Kharoof–Ipek–Okay, arXiv:2406.19961;
- confirmed through bibliographic records only: Penrose, *Structural Topology* 17 (1991);
- confirmed through secondary sources only: Aristotle, *Physics* III.6; Vorob'ev 1962; Howson 2008.

Files.

- Index: `glued_carriers_taxonomy.json`.
- Witness and prior art: `collision_vs_generation.md`, in the `delay_embedding` open questions.
- Lean (in the staging folder of the formalization):
  - rings: `WindowGluing.lean`;
  - triangle: `TriangleCyclic.lean`.
- Relation to the cycle side: `cyclic_marginal_scoping.md`.